achieved by reducing the gate-to-channel separation, which is an essential aspect of shrinking the transistor.

The devices exhibited excellent switching characteristics—switching is the essential function of a transistor—and high transconductance at low voltages. The researchers concluded that future CNFETs would probably outdo silicon transistors by an even larger margin.

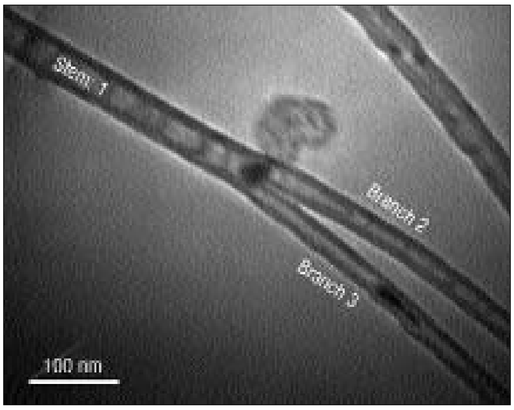
Beyond Just Transistors

Developments followed thick and fast. In early 2004, researchers at the University of California, Berkeley and Stanford University created "the first working, integrated silicon circuit that successfully incorporated carbon nanotubes in its design."

According to Jeffrey Bokor, principal investigator of the project, this was a first step in building "the most advanced nano-electronic products, in which (carbon nanotubes are placed) on top of a powerful silicon integrated circuit so that they can interface with an underlying information processing system." In other words, nanotubes had definitively entered the Integrated Circuit.

Nanotubes are naturally difficult to manipulate because of their size, and another groundbreaking announcement came in late 2004: NEC developed a way of positioning nanotubes, while also controlling their diameter. (It was an NEC researcher, if you recall, who discovered carbon nanotubes in soot.) NEC's goal was to develop chips running at 15 to 20 GHz (wow!) while consuming the same power as a Pentium 4—and this discovery was reported as an important step in that direction.

Not to be left behind, Intel announced in mid-2005 that it was stepping up its own research on nanotubes. Intel and other chipmakers will continue to use silicon for a long time, but their foray into nanotechnology—and nanotubes in particular—is indicative of how important this field of research is right now.



A photo of a Y-shaped nanotube that can act as a transistors, developed by researchers at UC San Diego and Clemson University. The voltage is applied to the stem

"Carbon nanotubes are already the top candidate to replace silicon when current chip features just can't be made any smaller

Dr Phaedon Avouris
IBM Fellow and manager of Nanometer Scale Science and Technology at the IBM T J Watson Research Center

The Plot Thickens

August 2005 saw the emergence of something entirely new: it turned out that nanotubes could *be* transistors! At the University of California in San Diego, Prab Bandaru and colleagues, and Apparao Rao (hoorah for Indians abroad!) of Clemson University, constructed Y-shaped nanotubes that could act as transistors. Applying a voltage to the stem of the Y precisely controlled the flow of electrons through the two branches. This is revolutionary when you think about the sizes: silicon transistors measure about a hundred nanometres today at the state of the art, and these Y-shaped nanotubes are just tens of nanometres large. It's almost unbelievable, actually: the transistors are fully self-contained, according to Bandaru.

In March of this year came an even more startling announcement: IBM researchers, working with researchers from the University of Florida and Columbia University, created *an integrated circuit out of a single carbon nanotube*. To be specific, they used an 18-micron long nanotube to build a 10-transistor ring oscillator, a type of circuit composed of an odd number of NOT gates. The IC is a million times faster than earlier ICs built using multiple nanotubes, but it clocks in at only 52 MHz—no GHz just yet!

But here's a dampener: in regard to commercialising nanotube chips, Fred Zieber, an analyst at Pathfinder Research, said: "It's a way off... It could be a few years or an eternity." But remember that this comment is about all-carbon chips, not about nanotubes being used in transistors: if you recall, transistors that use nanotubes can already outperform their silicon-only counterparts.

Looking Back

Nanotube research is on at various levels. Silicon and nanotube transistors can co-exist on the same substrate. Then there are efforts on to make nanotube transistors faster. The Y-shaped nanotubes that act as transistors are especially interesting, because that means no more silicon: and even more interesting is the recent development of an entire IC being constructed from a nanotube.

The major obstacles in all this are the manufacture of the nanotubes themselves—they come out as clumps of semi-conducting plus conducting types, and need to be filtered—and, of course, assembly, which is difficult because of the nanotubes' extremely small dimensions.

Looking Ahead

Things are moving fast: it was only proved in 2002 that nanotube transistors could outperform silicon ones, and in 2006, we're already seeing an entire IC made of a single nanotube! There are many years to go—Intel's roadmap, for example, contains several more generations of silicon. And it's only in 2018 that Moore's law—which seems to have envisaged only silicon—is supposed to die, as it were. Who can say what kind of research will happen in the next 12 years? We're prepared for year 2018, and carbon might take over even before that. Now that *is* a happy ending! ■

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